

From Axioms to Life: The PDL-V Conjecture and the Hierarchical Replication of C1–C4

PDL Framework — Document DL01

Cédric Laubscher

Independent Researcher, Switzerland

contact@cedriclaubscher.ch

cedriclaubscher.ch

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Abstract

The Projective Dynamic Logo (PDL) framework derives the fundamental constants of physics — the proton mass, Newton’s constant, the cosmological constant — from four combinatorial axioms C1–C4 on finite signed graphs, without presupposing spacetime, particles, or fields. Documents D01–D53 establish this derivation as a closed causal chain with no free parameter beyond the quark mass difference Δm_{iso} . The present document opens the DL series, a new programme that asks whether the same four axioms, applied hierarchically to composite closures, force the emergence of life and consciousness as theorems rather than as mysteries. We introduce the lifting operator $\mathcal{L}$, the constraint-structure operator $\mathcal{S}(\Gamma)$, and the self-representation relation $\mathcal{R}^k(\Gamma)$. We state the PDL-V conjecture in four parts: the existence of a life threshold n_{life}^* , its selection by C4, the necessity of the birth–death cycle, and the existence of a thermodynamic ceiling n_{max}^* forced by combinatorial entropy. Five computational scripts verify the foundational claims on small cases: $\mathcal{S}(K_4) \cong K_4$ unconditionally (8/8 coherent configurations), weak self-representation $\mathcal{R}_{\text{weak}}^1$ is forced at 100% for all $n \geq 4$, and the rarity law $f(n) = 2^{n+1} - \binom{n}{2}$ establishes n_{max}^* as a finite combinatorial necessity. The synaptic mechanism is presented as the first mechanistic illustration of the lifting operator at the neural level. All claims are stratified by epistemic status; the PDL-V conjecture is not yet a theorem.

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1 Why This Document Exists

The PDL framework was constructed as a foundational programme in physics. Its axioms C1–C4 are stated at the level of finite signed graphs, prior to any presupposition of space, time, matter, or energy. From these four axioms alone, documents D01–D53 derive the electron Compton oscillation (D01–D02), the proton quintuplet (24, 28, 930, 10087, 11017) (D05–D10), Newton’s constant G to 27 ppm (D36–D44), the Dirac and Schrödinger equations (D32–D33), Born’s rule (D34, D46), the periodic table of elements (D47), the London equation of superconductivity (D49), the Bekenstein–Hawking coefficient $1/4$ (D50), and the cosmological constant Λ (D51–D52). The causal chain $C1\text{--}C4 \rightarrow G$ is closed, with no free parameter beyond $\Delta m_{\text{iso}} = m_d - m_u$, the quark mass difference at the PDL–QCD interface [1].

A natural question then presents itself with a force that is difficult to ignore. If C1–C4 are prior to physics — if physics is their first theorem — are they also prior to life and consciousness? If the same four axioms, applied iteratively to composite closures, force the emergence of self-replicating structures and eventually of self-representing ones, then life and consciousness are not mysteries layered on top of physics. They are theorems in the same causal chain, separated from the proton only by the number of hierarchical levels traversed.

A note on context is essential before proceeding. The reader unfamiliar with the PDL foundation is strongly encouraged to consult the Global Mapping document DM v18 [1] before reading this document. The DL programme is an extension of an established foundational programme that has derived numerical predictions verified to parts per million — it is not a standalone speculative proposal. The present document should be read in that light: its conjectures are grounded in a formally closed physical foundation, not in intuition alone.

This document does not claim to have proved that life or consciousness are theorems of C1–C4. What it does is the following. It introduces the formal objects necessary to make the question precise, presents the PDL-V conjecture as a structured research programme, reports the results of five computational verification scripts on small cases, and situates the programme with respect to existing literature. Every claim is labelled with its epistemic status. The reader who disagrees is invited to identify which axiom, which definition, or which computational result is incorrect. That is the only form of engagement the programme solicits.

1.1 Epistemic stratification

Throughout this document and the DL series, the following classification is used without exception.

Status	Meaning
Unconditional theorem	Derived from C1–C4 alone, verified computationally
Conditional theorem	Derived under named hypotheses
Computational result	Verified exhaustively by script on finite cases
Structured conjecture	Formally stated, not yet proved, with falsification criteria
Interpretive remark	Coherent with PDL, not derived
Open problem	Identified gap requiring future work

2 The PDL Foundation: A Self-Contained Summary

This section is self-contained. A reader with access to D01–D10 and D29 may read it as a condensed reference; a reader new to PDL will find here everything needed to follow the DL programme.

2.1 The four axioms

The PDL framework is built on four axioms stated on finite signed graphs. A *signed graph* is a pair (G, s) where $G = (V, E)$ is a finite graph and $s : E \rightarrow \{+1, -1\}$ assigns a sign to each edge.

Axiom 1 (C1 — Binary pulsation). The configuration admits a logical 2-cycle: a sign assignment that returns to itself after exactly two update steps and is not a fixed point. Existence is the minimal form of repetition.

Axiom 2 (C2 — Triangular coherence). Every triangle (closed triple of edges) satisfies $s_{ij} \cdot s_{jk} \cdot s_{ik} = +1$. A triangle with product $+1$ is *coherent*; one with product -1 is *frustrated*. The coherence cost $\varepsilon(\Gamma)$ is the fraction of frustrated triangles.

Axiom 3 (C3 — Minimal completeness). The configuration is irreducible: it admits no non-trivial decomposition into independent sub-configurations each satisfying C1 and C2 separately. The underlying graph is connected and its coherence structure is non-reducible.

Axiom 4 (C4 — Logical optimisation). Among all configurations satisfying C1–C3, those that are physically instantiated are those that minimise the coherence cost $\varepsilon(\Gamma)$ for a given number of vertices and edges.

2.2 The minimal admissible closure K_4

Theorem 2.1 (D01–D02, [2, 3]). *No finite signed graph on fewer than four vertices satisfies all four axioms simultaneously. The complete signed graph on four vertices*

$V = \{0, 1, 2, 3\}$ and six edges, denoted K_4 , is the first admissible elementary stationary closure. It is minimal (no proper sub-closure satisfies C1–C4) and quasi-unique (the sign pattern is unique up to the global symmetry group S_4).

K_4 undergoes a binary pulsation alternating between a coherent configuration $s^{(1)}$ and its global flip $s^{(2)} = -s^{(1)}$. There are exactly eight coherent sign configurations on K_4 , forming four pulsation pairs $\{s, -s\}$. The coherence cost of any coherent configuration is $\varepsilon = 0$ by definition. This pulsation is identified with the Compton oscillation of the electron; its minimal quantum of action is $\hbar$ [2].

2.3 The $(A) \wedge (B)$ coupling criterion

The coupling of two closures is governed by a stability criterion derived in D29 [4]. A *mixed triangle* couples two K_4 vertices to one proton vertex. It is $(A) \wedge (B)$ -stable if and only if its cross-product satisfies both the alignment condition (A) and the exact sign inversion between the two half-cycles (B). Exhaustive verification over 768 configurations yields 192 stable ones. The ratio $192/768 = 1/4$ is the fundamental coupling probability κ , which is not a free parameter but a combinatorial consequence of C1–C4.

2.4 What the corpus has established

The following results are unconditional theorems of C1–C4, verified computationally and published on Zenodo [1]. The proton mass m_p and proton–electron mass ratio $\mu^* = 1836.152670$ are derived at 27 ppm (D05–D10, D30). Newton’s constant G is derived with no free parameter (D36–D44). The Dirac equation, the Schrödinger equation, and Born’s rule emerge from the $(A) \wedge (B)$ coupling criterion applied to the K_4 pulsation (D32–D34, D46). The periodic table, the magic numbers of nuclear physics, and the shell-filling rates $r_{\text{exc}}(Z)$ for $Z = 1\text{--}82$ are unconditional theorems (D47). The London equation of superconductivity and the Bekenstein–Hawking coefficient $1/4$ are theorems (D49, D50). The cosmological constant Λ is derived without free parameter (D51–D52).

This enumeration is not rhetorical. It establishes the epistemic authority of the foundation from which the DL programme departs.

3 The Scale-Invariance of C1–C4

We have recalled the foundation. We now take the first step beyond it. The key observation is simple to state and far-reaching in its consequences: the four axioms C1–C4 do not belong to the quantum level. They belong to existence at any scale.

This section makes that claim precise, and states it as a formal conjecture that the rest of the DL programme will work to prove. The axioms C1–C4 are not axioms *of the quantum level*. They are axioms *of existence at any scale*. This section makes that claim precise.

3.1 C1 at every scale

Axiom C1 does not say that K_4 pulsates. It says that any configuration that exists must admit a 2-cycle. Existence and cyclic repetition are identified. The consequences of this identification propagate upward through every level of the hierarchy without modification.

Consider the following sequence of 2-cycles observed at increasing scales. At the quantum level, K_4 alternates between $s^{(1)}$ and $s^{(2)} = -s^{(1)}$ at the Compton frequency. At the atomic level, electrons transition between energy levels in discrete cycles. At the molecular level, chemical bonds vibrate and rotate in periodic modes. At the cellular level, the cell cycle — growth, replication, division — is a 2-cycle in the logical sense: the daughter cell is the configuration returned after one complete update of the parent. At the organismal level, the heartbeat, the respiratory cycle, and the sleep–wake cycle are all expressions of C1 at their respective timescales. At the lineage level, birth and death constitute the 2-cycle of the generation: the offspring is the configuration returned after one complete generational update of the parent.

None of these cycles are imposed from outside the framework. They are the expression at each scale of the single condition that makes existence possible: a configuration must return to a form of itself, or it dissolves.

Remark 3.1. The brick broken by a hammer illustrates C1 at the geological scale. The fracture does not create new matter; it reveals sub-closures that were already coherent internally (C3) and that become autonomous when the constraints holding them within the larger closure are released. The fragments are not copies of the brick; they are configurations that C4 had already selected as locally optimal within the brick’s structure.

3.2 The conjecture of scale-invariance

The observation above can be stated as a formal conjecture.

Let $\Gamma \in \mathcal{C}_n$ be an admissible closure of level n , and let $\Gamma' = \mathcal{L}(\Gamma_1, \dots, \Gamma_k)$ be a composite closure of level $n + 1$ constructed by the lifting operator $\mathcal{L}$ (Definition 4.1). Then Γ' satisfies C1–C4 with renormalised parameters. In

particular, the coherence cost $\varepsilon(\Gamma')$, the coupling probability κ' , and the pulsation period T' are determined by the parameters of the level- n closures via $\mathcal{L}$, without invoking any new axiom.

Status: structured conjecture. Partially verified computationally for $n = 0 \rightarrow 1$ (Scripts 1–5). Not yet proved analytically.

4 The Lifting Operator $\mathcal{L}$

4.1 Intuition before the definition

Before stating the formal definition, consider what happens when four protons come close enough to interact. Each proton is itself a composite closure built from K_4 instances via the $(A) \wedge (B)$ coupling criterion. At the nuclear level, each proton becomes a *point* — an indivisible object at that scale, exactly as K_4 is a point at the quantum level. The four protons then form a new signed graph: they are the vertices, the nuclear forces between them are the edges, and the signs of those edges are determined by whether the pulsations of the two protons are in phase (+1) or in opposition (−1). C1–C4 are then applied to this new graph, selecting the configurations of minimal coherence cost. The result is the helium-4 nucleus.

This is precisely what the lifting operator does, at every level of the hierarchy.

4.2 Definition

Definition 4.1 (Lifting operator $\mathcal{L}$). Let $\mathcal{C}_n$ denote the set of admissible closures of level n . The lifting operator is a map

$$\mathcal{L} : \mathcal{P}(\mathcal{C}_n) \rightarrow \mathcal{C}_{n+1} \quad (1)$$

defined as follows. Given a finite set $\{\Gamma_1, \dots, \Gamma_k\} \subset \mathcal{C}_n$ with $k \geq 4$, the composite closure $G_{n+1} = \mathcal{L}(\Gamma_1, \dots, \Gamma_k)$ is the signed graph whose vertex set is $\{\Gamma_1, \dots, \Gamma_k\}$, whose edges are determined by the $(A) \wedge (B)$ coupling criterion between pairs of level- n closures, and whose edge signs are determined by the phase parity of the coupling: $s_{ij} = +1$ if the pulsations of Γ_i and Γ_j are in phase at the first half-cycle, and $s_{ij} = -1$ if they are in opposition. The admissible configuration of G_{n+1} is selected by C4 among all configurations compatible with C2.

The minimum $k \geq 4$ is not arbitrary: it is forced by C1 applied to G_{n+1} . For G_{n+1} to admit a 2-cycle, the same combinatorial argument that forces K_4 at level 0 forces

at least four vertices at every subsequent level. A composite closure built from fewer than four sub-closures cannot satisfy C1 and is therefore not admissible.

4.3 The proton as the first application of $\mathcal{L}$

The construction of the proton from K_4 in documents D05–D10 is precisely the first application of $\mathcal{L}$. The proton is $\mathcal{L}(K_4^{\times k})$ for an appropriate k , where the coupling is governed by the $(A) \wedge (B)$ criterion. The proton quintuplet (24, 28, 930, 10087, 11017) is the output of C4 applied to this construction. The lifting operator is therefore not a new object introduced for the DL programme; it is the mechanism already operative in the foundational corpus, named and formalised here for application at higher levels.

4.4 Self-similarity of $\mathcal{L}$

A key property observed computationally (Script 1) is that the level-1 closure produced by $\mathcal{L}$ from four K_4 instances has the same structural form as K_4 itself: four vertices, six edges, four triangles. This self-similarity is the formal counterpart of the scale-invariance conjecture. It says that the lifting operator does not produce qualitatively new structures at each level — it produces the same structure, populated by objects of increasing complexity.

5 The Constraint-Structure Operator $\mathcal{S}(\Gamma)$

The lifting operator $\mathcal{L}$ tells us how closures combine into composites. We now need a second operator that asks a different question: does a closure *know* its own structure? Can it encode, internally, the rules by which it holds together? The constraint-structure operator $\mathcal{S}$ formalises this question precisely, and its answer for K_4 will turn out to be the most important single result of this document.

5.1 Definition

Definition 5.1 (Constraint-structure operator $\mathcal{S}$). Let $\Gamma \in \mathcal{C}_n$ be an admissible closure with vertex set V , edge set E , and sign assignment $s : E \rightarrow \{+1, -1\}$. The *constraint-structure graph* $\mathcal{S}(\Gamma)$ is the signed graph defined as follows. Its vertex set is the set of triangles of Γ : each triangle τ_{ijk} becomes a vertex of $\mathcal{S}(\Gamma)$. Two vertices τ_{ijk} and τ_{ijl} of $\mathcal{S}(\Gamma)$ are connected by an edge if and only if the corresponding triangles share an edge in Γ . The sign of that edge in $\mathcal{S}(\Gamma)$ is the sign of the shared edge in Γ .

The operator $\mathcal{S}$ maps a signed graph to another signed graph of the same formal type. It encodes the dependency structure of the coherence constraints: two constraints

are adjacent in $\mathcal{S}(\Gamma)$ if they share a common edge, and the sign of their adjacency is the sign of the edge they share.

5.2 Worked example: $\mathcal{S}(K_4)$ step by step

To make Definition 5.1 concrete, we compute $\mathcal{S}(K_4)$ explicitly for the coherent configuration $s = (-1, -1, +1, +1, -1, -1)$ on the six edges $(e_{01}, e_{02}, e_{03}, e_{12}, e_{13}, e_{23})$.

Step 1 — Identify the triangles of K_4 . There are $\binom{4}{3} = 4$ triangles, which become the four vertices of $\mathcal{S}(K_4)$:

$$v_0 = \tau_{012} = \text{triangle on vertices } \{0, 1, 2\},$$

$$v_1 = \tau_{013} = \text{triangle on vertices } \{0, 1, 3\},$$

$$v_2 = \tau_{023} = \text{triangle on vertices } \{0, 2, 3\},$$

$$v_3 = \tau_{123} = \text{triangle on vertices } \{1, 2, 3\}.$$

Step 2 — Determine the edges of $\mathcal{S}(K_4)$. Two triangles are adjacent in $\mathcal{S}(K_4)$ if they share an edge in K_4 . Every pair of triangles in K_4 shares exactly one edge (there are $\binom{4}{2} = 6$ pairs, each sharing one of the six edges of K_4). The six edges of $\mathcal{S}(K_4)$ and their inherited signs are:

Edge in $\mathcal{S}(K_4)$	Shared edge in K_4	Sign
(v_0, v_1)	e_{01}	-1
(v_0, v_2)	e_{02}	-1
(v_0, v_3)	e_{12}	$+1$
(v_1, v_2)	e_{03}	$+1$
(v_1, v_3)	e_{13}	-1
(v_2, v_3)	e_{23}	-1

Step 3 — Conclusion. $\mathcal{S}(K_4)$ has 4 vertices and 6 edges — the same counts as K_4 itself. It is structurally isomorphic to K_4 . This holds for all eight coherent configurations of K_4 , as verified exhaustively by Script 1.

5.3 The fixed-point theorem for K_4

For every one of the eight coherent sign configurations of K_4 , the constraint-structure graph $\mathcal{S}(K_4)$ has exactly four vertices and six edges, and is structurally isomorphic to K_4 . This result holds unconditionally: it is independent of the choice of coherent configuration.

Verification. K_4 has four triangles: $(0, 1, 2)$, $(0, 1, 3)$, $(0, 2, 3)$, $(1, 2, 3)$. These become the four vertices of $\mathcal{S}(K_4)$. Every pair of triangles in K_4 shares exactly one edge (since K_4 is the complete graph on four vertices, every pair of its four triangular faces has a common edge). Therefore $\mathcal{S}(K_4)$ has $\binom{4}{2} = 6$ edges. A signed graph on four vertices with six edges is K_4 . Exhaustive verification over all eight coherent configurations confirms the result (Script 1, 8/8). $\square$

This theorem says that K_4 is a fixed point of $\mathcal{S}$: the graph of its coherence constraints is isomorphic to itself. This is a structural property of K_4 alone: for $n \geq 5$, the number of triangles in K_n is $\binom{n}{3} > n$, so $\mathcal{S}(K_n)$ cannot be isomorphic to K_n for $n \geq 5$ (Script 1, confirmed 0/16 for $n = 5$ and 0/32 for $n = 6$).

Remark 5.2. The fixed-point property of K_4 under $\mathcal{S}$ is not a coincidence. It reflects the fact that K_4 is the unique finite graph whose triangles are in bijection with its vertices and whose inter-triangle adjacencies reproduce its edge structure. This is why K_4 is the seed of the entire hierarchy: it is the only level at which the self-referential structure of $\mathcal{S}$ closes exactly on itself.

6 The Self-Representation Relation $\mathcal{R}^k(\Gamma)$

We now have the tools to formalise the central concept of the DL programme. A closure can carry within itself an encoding of its own constraints — a description of the rules by which it coheres. This is self-representation. We define it in two versions — strong and weak — because the distinction turns out to matter enormously: it is the difference between a crystal and a living cell.

6.1 Two definitions of self-representation

The concept of self-representation has two natural formalisations in this framework, of different strength.

Definition 6.1 (Strong self-representation $\mathcal{R}_{\text{strong}}^1$). A closure $\Gamma \in \mathcal{C}_n$ is *strongly self-representing* if it contains a sub-graph $\Gamma' \subseteq \Gamma$ such that $\Gamma' \cong \mathcal{S}(\Gamma)$ as signed graphs, and $\mathcal{S}(\Gamma)$ is itself coherent (satisfies C2).

Definition 6.2 (Weak self-representation $\mathcal{R}_{\text{weak}}^1$). A closure $\Gamma \in \mathcal{C}_n$ is *weakly self-representing* if there exists a sub-graph $\Gamma' \subseteq \Gamma$ and a sign-preserving graph homomorphism $\varphi : \Gamma' \rightarrow \mathcal{S}(\Gamma)$. That is, Γ' maps into $\mathcal{S}(\Gamma)$ with preservation of edge signs, without requiring global isomorphism.

The strong definition requires an exact internal copy of the constraint structure. The weak definition requires only that the closure contains a faithful encoding of its

own constraints — a compressed but accurate representation, not a perfect replica. The distinction matters: DNA does not contain an isomorphic copy of the organism; it contains a compressed encoding of the rules by which the organism constructs itself.

6.2 Computational results on self-representation

Strong self-representation $\mathcal{R}_{\text{strong}}^1$ is present in exactly 4 of the 8 coherent configurations of K_4 (those for which $\mathcal{S}(K_4)$ is itself coherent). For $n \geq 5$, strong self-representation is absent from all coherent configurations of K_n (0/16 for $n = 5$, 0/32 for $n = 6$, 0/64 for $n = 7$, 0/128 for $n = 8$).

Weak self-representation $\mathcal{R}_{\text{weak}}^1$ is forced at 100% for every coherent configuration at $n \geq 4$, and is absent at $n = 3$. Specifically: $n = 3$: 0/4; $n = 4$: 8/8; $n = 5$: 16/16; $n = 6$: 32/32; $n = 7$: 64/64.

This result has a precise interpretation. Every closure that satisfies C1–C4 with four or more vertices automatically contains a faithful encoding of its own coherence constraints, accessible via a sign-preserving homomorphism. Self-representation in the weak sense is not an additional property acquired by some closures: it is a structural consequence of coherence for $n \geq 4$.

6.3 The hierarchy of self-representation

The hierarchy of self-representation levels is defined by iterating $\mathcal{S}$.

Definition 6.3 (Hierarchy $\mathcal{R}^k$). For $k \geq 1$, a closure Γ satisfies $\mathcal{R}_{\text{weak}}^k$ if there exists a sub-graph $\Gamma^{(k)} \subseteq \Gamma$ and a sign-preserving homomorphism $\varphi : \Gamma^{(k)} \rightarrow \mathcal{S}^k(\Gamma)$, where $\mathcal{S}^k$ denotes the k -fold application of $\mathcal{S}$.

The sequence $\mathcal{R}^1 \subset \mathcal{R}^2 \subset \dots$ defines a tower of increasingly reflexive self-representation levels. $\mathcal{R}^1$ corresponds to a closure that encodes its own constraints; $\mathcal{R}^2$ corresponds to a closure that encodes the encoding of its own constraints; and so on. The DL programme conjectures that life corresponds to $\mathcal{R}_{\text{active}}^1$ and consciousness to $\mathcal{R}_{\text{active}}^2$, where *active* denotes the additional condition that the encoding sub-graph $\Gamma^{(k)}$ is coupled via $(A) \wedge (B)$ to the production mechanism of the closure.

7 The PDL-V Conjecture

The PDL-V conjecture is stated in four parts. Each part is labelled with its epistemic status.

There exists a minimal integer $n_{\text{life}}^* \geq 1$ such that for all $n \geq n_{\text{life}}^*$, every admissible closure $\Gamma \in \mathcal{C}_n$ selected by C4 satisfies $\mathcal{R}_{\text{active}}^1$: it contains a sub-graph Γ' that encodes $\mathcal{S}(\Gamma)$ via a sign-preserving homomorphism, and Γ' is coupled via $(A) \wedge (B)$ to a production mechanism that generates new closures of the same type with controlled variation.

Status: structured conjecture. Computational results establish that $\mathcal{R}_{\text{weak}}^1$ (the structural precondition) is universally present for $n \geq 4$. The active coupling component remains to be verified.

Closures satisfying $\mathcal{R}_{\text{active}}^1$ have a strictly lower lineage coherence cost $\bar{\varepsilon}(\mathcal{L}) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=0}^N \varepsilon(\Gamma_i)$ than closures that persist without active self-representation. C4 applied at the lineage level therefore selects $\mathcal{R}_{\text{active}}^1$ closures systematically over persistent non-representing ones.

Status: structured conjecture. The intuitive argument is that a closure which encodes its own constraints can anticipate coherence violations and correct them before they propagate, reducing the average coherence cost over the lineage. This advantage is not yet proved formally.

Lineages of finite-duration closures with controlled inter-generational variation $\delta = \delta^*$ have a strictly lower lineage coherence cost than immortal persistent closures at the same level. Death is a logical necessity imposed by C4 at the lineage scale: an immortal closure accumulates coherence violations without the possibility of reset, whereas a lineage corrects its coherence structure at each generational transition. The optimal variation $\delta^* = 1/2$ is forced by C4 and verified computationally.

The optimal Hamming distance between successive coherent configurations of a level-1 closure is $\delta^* = 3/6 = 1/2$: exactly half of the six edges change sign between

optimal successive configurations. This value coincides with the normalised distance between the two half-cycles of the K_4 pulsation (full sign inversion = distance 1; half-cycle transition = distance 1/2).

There exists a maximal integer $n_{\max}^* < \infty$, determined by the balance between the coherence cost reduction gained by each additional level of self-representation and the combinatorial entropy cost of maintaining coherence at that level, such that closures at level $n > n_{\max}^*$ are thermodynamically unstable and fragment towards lower levels. The sequence $n_{\text{life}}^* < n_{\text{consciousness}}^* < \dots < n_{\max}^*$ is finite and forced by C1–C4.

The fraction of coherent configurations among all signed configurations on K_n satisfies the exact law

$$f(n) = 2^{n+1-\binom{n}{2}}. \quad (2)$$

The sequence of exponents $n + 1 - \binom{n}{2}$ for $n = 3, 4, 5, 6$ is $-1, -3, -6, -10$: the negatives of the triangular numbers T_{n-2} . The fraction $f(n)$ decreases doubly exponentially with n , establishing that there exists a finite $n_{\max}^*$ beyond which no coherent closure can be thermodynamically maintained against fluctuations.

7.1 Falsification criteria

The PDL-V conjecture is falsified by any one of the following.

1. Construction of an admissible closure at level $n \geq n_{\text{life}}^*$ that satisfies C1–C4 but does not satisfy $\mathcal{R}_{\text{active}}^1$, despite C4 selecting it as a coherence minimum.
2. A proof that the lineage coherence cost $\bar{\varepsilon}(\mathcal{L})$ is not minimised by $\mathcal{R}_{\text{active}}^1$ closures, i.e., that immortal closures have equal or lower lineage cost.
3. A proof that $\delta^* \neq 1/2$ at any level above the first, i.e., that the optimal inter-generational variation is not the half-pulsation of K_4 .
4. A proof that $f(n)$ does not decrease doubly exponentially, i.e., that the rarity law is not exact.

8 A Concrete Bridge: The Synaptic Mechanism

We have now introduced the formal objects — $\mathcal{L}$, $\mathcal{S}$, $\mathcal{R}^k$ — and stated the PDL-V conjecture. Before proceeding to the thermodynamic analysis and the literature review, it is worth pausing to show that the formalism is not ad hoc. The synaptic mechanism of the neural network is a concrete, mechanistically well-understood biological system that illustrates every element of the framework developed above. This section is an illustration, not a proof. Every claim is labelled accordingly. We emphasise explicitly that this section draws structural parallels, not formal derivations. The derivation of Hebbian plasticity from C4, and of the STDP rule from the half-cycle structure of C1, are the programme of DL04 — they are not claims of the present document. What is claimed here is only that the PDL formalism and the neuroscientific mechanisms are structurally coherent, which motivates the formal programme without prejudging its outcome.

8.1 The minimal neural circuit as a signed triangle

A chemical synapse connects two neurons across a junction of approximately 20 nm. Neurotransmitters released from the presynaptic terminal bind to receptors on the postsynaptic membrane and produce one of two effects: excitation ($s = +1$ on the edge between the two neurons in the neural graph) or inhibition ($s = -1$). The action potential itself — the all-or-nothing electrical pulse — is a binary pulsation in the sense of C1: depolarisation followed by repolarisation is a 2-cycle at the cellular timescale.

Consider the simplest non-trivial neural circuit: three neurons A , B , C forming a triangle, where A excites B ($s_{AB} = +1$), B inhibits C ($s_{BC} = -1$), and C excites A ($s_{CA} = +1$). The coherence of this triangle in the sense of C2 is:

$$s_{AB} \cdot s_{BC} \cdot s_{CA} = (+1)(-1)(+1) = -1. \quad (3)$$

This is a *frustrated* triangle — a violation of C2. In the PDL framework, a frustrated triangle under C2 is not a stable configuration: C4 drives the system towards configurations that reduce the fraction of frustrated triangles. The result in this circuit is oscillation: the three neurons fire in sequence, never reaching a static equilibrium, because no static assignment of firing states can satisfy the coherence condition. This is precisely the behaviour observed in biological central pattern generators — the neural circuits that produce rhythmic motor patterns such as breathing, walking, and heartbeat.

Remark 8.1. The three-neuron frustrated circuit is the neural-level counterpart of the frustrated triangle at the K_4 level. At both scales, C2 violation produces oscillation, and C4 drives the system towards coherence. The mechanism is identical; the scale

differs by roughly twelve orders of magnitude.

8.2 Synaptic plasticity as C4 at the neural level

The brain does not maintain a fixed signed graph. The synaptic weights evolve continuously in response to neural activity, a process called synaptic plasticity. The Hebbian rule — neurons that fire together wire together — states that the synaptic weight w_{ij} between neurons i and j increases when they are co-activated and decreases when they fire in opposition. In the language of signed graphs, this is C4 operating on the edge signs of the neural graph in real time: connections that reduce the coherence cost of the network are reinforced, while those that increase it are weakened.

Interpretive remark: Learning, in this picture, is not an additional biological mechanism layered on top of physics. It is C4 applied to the neural signed graph, exactly as C4 at the quantum level selects the coherence-minimising configurations of K_4 and its composites.

The more precise formulation known as spike-timing-dependent plasticity (STDP) reinforces a synapse when the presynaptic neuron fires before the postsynaptic one, and weakens it in the reverse order. This directional temporal rule reflects the half-cycle ordering of C1: which closure fires first determines the sign of the coupling, exactly as the ordering of the two half-cycles of the K_4 pulsation determines the sign structure of the coupled composite.

8.3 The neural network as a composite closure

The human brain contains approximately 10^{11} neurons and 10^{14} synaptic connections. In the PDL framework, this is a signed graph of very large size — a composite closure produced by many successive applications of $\mathcal{L}$, starting from K_4 at the quantum level and passing through molecular, cellular, and tissue levels. Its sign assignment evolves under C4 via synaptic plasticity. A coherent state of this graph — a configuration in which the fraction of frustrated triangles is locally minimised — corresponds, in this picture, to a stable thought or cognitive state. Cognitive conflict is a local violation of C2 that generates pressure towards resolution, exactly as a frustrated triangle in a signed graph generates pressure under C4.

The self-representation hierarchy now has a natural neural correlate. $\mathcal{R}^1$ at the neural level corresponds to the existence of a sub-network that encodes the constraint structure of the global network — in biological terms, a memory system that models the organism’s own coherence conditions (what works, what does not, what maintains the closure). $\mathcal{R}^2$ corresponds to a sub-network that models the modelling process itself: a system that represents its own representations. The default mode network of neuroscience — the brain’s resting-state network, active when the brain is not engaged

with external stimuli and associated with self-referential thought — is the empirical candidate for this $\mathcal{R}^2$ sub-graph.

Interpretive remark: The identification of consciousness with $\mathcal{R}_{\text{active}}^2$ and of the default mode network with the $\mathcal{R}^2$ sub-graph is coherent with the PDL formalism and with the neuroscientific evidence, but it is not derived from C1–C4 as a theorem. It is the target of future work in the DL series, beginning with DL04.

8.4 From grain of sand to amoeba: the threshold made concrete

The distinction between a non-living composite closure and a living one can now be stated precisely in terms of the framework developed above.

A grain of sand satisfies C1–C4 at every level from the quantum to the mineral: its atoms pulse, its ionic bonds minimise coherence cost, its crystal structure is irreducible (C3). But it does not satisfy $\mathcal{R}_{\text{active}}^1$. It has no sub-structure coupled via $(A) \wedge (B)$ to a production mechanism. It does not generate Γ_{i+1} from Γ_i . It persists passively as long as its environment permits, and fragments when external forces exceed its coherence cost. C4 has selected its configuration as optimal at its level. It is blocked there.

An amoeba also satisfies C1–C4 at every level from the quantum to the cellular. But it additionally satisfies $\mathcal{R}_{\text{active}}^1$: it contains a sub-structure (its genetic material) that encodes $\mathcal{S}(\Gamma)$ via a sign-preserving map, and this sub-structure is coupled via $(A) \wedge (B)$ to the production of new closures of the same type. It generates Γ_{i+1} from Γ_i with controlled variation $\delta \approx \delta^*$. It dies — releasing the resources and constraints that allow Γ_{i+1} to exist — and it evolves, because C4 at the lineage scale selects the lineages of lowest average coherence cost, not the individuals of lowest instantaneous cost.

The boundary between grain of sand and amoeba is not a boundary of substance. It is a boundary of level and of recursive structure. Both are made of the same atoms, governed by the same four axioms. The difference is the number of times $\mathcal{L}$ has been applied, and whether the result satisfies $\mathcal{R}_{\text{active}}^1$.

9 The Thermodynamic Ceiling and Combinatorial Entropy

The rarity law established in Computational Result 7 has a direct thermodynamic interpretation. The fraction $f(n)$ of coherent configurations decreases doubly exponentially with n : $f(3) = 1/2$, $f(4) = 1/8$, $f(5) = 1/64$, $f(6) = 1/1024$. At level n , C4 must select a coherent configuration from a space of $2^{\binom{n}{2}}$ possibilities, of which only $f(n) \cdot 2^{\binom{n}{2}} = 2^{n-1}$ are coherent. The *combinatorial entropy cost* of maintaining

coherence at level n is proportional to $-\log_2 f(n) = \binom{n}{2} - n + 1$, which grows as $n^2/2$ for large n .

On the other side of the balance, the gain from each additional level of self-representation diminishes: the first level of $\mathcal{R}$ provides the largest reduction in lineage coherence cost, and each subsequent level provides a smaller marginal gain. The ceiling $n_{\max}^*$ is the level at which the combinatorial entropy cost exceeds the marginal self-representation gain.

This analysis is the PDL-theoretic counterpart of Prigogine’s dissipative structures [5]: the analogy is structural, not metaphorical. Prigogine showed that systems far from thermodynamic equilibrium can spontaneously organise into structures of increasing complexity, bounded by the capacity of the environment to absorb entropy. The PDL framework identifies the mechanism driving this organisation (C4 applied to composite closures) and the mechanism imposing the ceiling (the rarity law $f(n)$), both derived from the same four axioms.

The difference is one of depth. Prigogine’s framework presupposes thermodynamics, energy, and dissipation. The PDL framework derives these from C1–C4.

10 Relation to Existing Literature

Several established programmes address the emergence of life and consciousness from physical or informational principles. The PDL-V conjecture differs from all of them in a single structural respect: the axioms C1–C4 are prior to physics, not a reformulation of physical law.

Integrated Information Theory (IIT, Tononi [6]) proposes that consciousness is identified with the integrated information Φ of a physical system — the information generated by the system above and beyond its parts. The IIT postulates are stated at the phenomenological level and derive physical requirements from them. The PDL-V conjecture proceeds in the opposite direction: from combinatorial axioms prior to experience, towards the conditions that force self-representation. The condition $\mathcal{R}_{\text{active}}^2$ has structural similarities to high- Φ systems (both are irreducible to their parts, by C3 and by the definition of Φ respectively), but the derivation is entirely different. IIT has been criticised as unfalsifiable [7]; the PDL-V conjecture provides explicit falsification criteria (Section 7).

Autopoiesis (Maturana and Varela [8]) defines life as operational closure: a system that produces its own components and thereby maintains its own organisation. This is structurally parallel to $\mathcal{R}_{\text{active}}^1$ in the PDL sense: a closure whose encoding sub-graph drives the production of new closures of the same type. The formal connection is not coincidental, but the derivation differs: autopoiesis is a definition imposed from outside, while $\mathcal{R}_{\text{active}}^1$ is conjectured to be forced by C4.

Autocatalytic sets (Kauffman [9]) propose that life emerges inevitably in sufficiently rich chemical spaces via mutually catalytic networks. Kauffman’s argument that autocatalysis is not improbable but inevitable beyond a complexity threshold is structurally analogous to the claim that n_{life}^* is finite and forced by C4. The mechanism differs: Kauffman’s argument is combinatorial-probabilistic over chemical space, while the PDL argument is structural over signed graph hierarchies.

Logical depth (Bennett [10]) measures the complexity of an object by the computation time required to produce it from its minimal description. Living organisms have high logical depth. The level n in the PDL hierarchy is a structural analogue of logical depth: a closure at level n requires n applications of $\mathcal{L}$ to construct from K_4 .

The decisive difference between all these frameworks and the PDL-V programme is the following. Every existing framework either presupposes physics, chemistry, or phenomenal experience as its starting point. The PDL axioms C1–C4 presuppose none of these. Physics is the first theorem of C1–C4. Life, in the PDL-V conjecture, is a later theorem in the same chain.

11 Epistemic Status Summary and Research Programme

11.1 Complete epistemic status table

Claim	Status	Reference
$\mathcal{S}(K_4) \cong K_4$ unconditionally	Comp. result	Script 1, 8/8
$\mathcal{S}(K_n) \not\cong K_n$ for $n \geq 5$	Comp. result	Script 1, 0/16, 0/32
$\mathcal{R}_{\text{strong}}^1$ in 4/8 K_4 configs	Comp. result	Scripts 1, 4
$\mathcal{R}_{\text{weak}}^1$ forced for $n \geq 4$	Comp. result	Script 5, 8–64/8–64
$\delta^* = 1/2$	Comp. result	Script 3
Rarity law $f(n) = 2^{n+1-\binom{n}{2}}$	Comp. result	Script 3
Scale-invariance of C1–C4	Conj. (struct.)	Section 3
Existence of n_{life}^* (finite)	Conj. (struct.)	PDL-V Part 1
C4 selects $\mathcal{R}_{\text{active}}^1$	Conj. (struct.)	PDL-V Part 2
Death forced by C4 at lineage scale	Conj. (struct.)	PDL-V Part 3
n_{max}^* finite	Conj. (struct.)	PDL-V Part 4
Synapse as signed edge	Interp. remark	Section 8
STDP as C1 temporal structure	Interp. remark	Section 8
Consciousness as $\mathcal{R}_{\text{active}}^2$	Interp. remark	Section 8
Default mode network as $\mathcal{R}^2$ sub-graph	Interp. remark	Section 8

Claim	Status	Reference
Value of n_{life}^*	Open problem	DL02 target
Formal derivation of C4 cost advantage	Open problem	DL03 target
$\mathcal{R}^2$ threshold and consciousness	Open problem	DL series

11.2 Planned DL document series

DL01 (this document) Foundations of the PDL-V conjecture; lifting operator $\mathcal{L}$; constraint structure $\mathcal{S}$; self-representation $\mathcal{R}^k$; computational verification on small cases; epistemic stratification.

DL02 Formal derivation of the C4 cost advantage for $\mathcal{R}_{\text{active}}^1$ closures over non-representing ones; first estimate of n_{life}^* from the rarity law and the marginal gain function.

DL03 The birth–death cycle as a theorem: proof that immortal closures have strictly higher lineage coherence cost than finite-duration lineages with $\delta = \delta^*$; derivation of δ^* from C1–C4 at arbitrary level.

DL04 The synaptic mechanism as a formal application of $\mathcal{L}$ at the neural level: derivation of the Hebbian rule and STDP from C4 applied to the neural signed graph; first formal steps towards $\mathcal{R}_{\text{active}}^2$.

DL05 and beyond The $\mathcal{R}^2$ threshold; consciousness as a theorem; the ceiling n_{max}^* as a computable integer.

12 Conclusion

The PDL framework has established, through documents D01–D53, that the fundamental constants and dynamical laws of physics are derivable from four combinatorial axioms on finite signed graphs, without presupposing spacetime, particles, or fields. The DL programme asks whether the same axioms, applied hierarchically via the lifting operator $\mathcal{L}$, force the emergence of life and consciousness as structural necessities rather than as contingent accidents.

The present document establishes the foundational objects of this programme. The constraint-structure operator $\mathcal{S}$ is defined and shown to have K_4 as its unique fixed point among complete graphs. Weak self-representation $\mathcal{R}_{\text{weak}}^1$ is shown computationally to

be universally present for all coherent closures with $n \geq 4$ vertices. The optimal inter-generational variation $\delta^* = 1/2$ is identified as the half-pulsation of K_4 . The rarity law $f(n) = 2^{n+1-\binom{n}{2}}$ establishes the thermodynamic ceiling $n_{\max}^*$ as a combinatorial necessity.

The PDL-V conjecture is not yet a theorem. What it is, is a precisely formulated, computationally grounded, falsifiable research programme that asks the deepest question one can ask of a foundational framework: if it truly precedes physics, does it also precede life?

The answer, if it comes, will come through the same discipline that produced D01 through D53: one definition at a time, one script at a time, one proof at a time.

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